

Research Proposal

Title: Blind Spot Reduction in Gastric Endoscopy with YOLO and Deep Learning

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1. Introduction

Gastrointestinal endoscopy is the gold standard for diagnosing gastric diseases. However, diagnostic accuracy heavily depends on the endoscopist's experience, which requires years of training. Studies have shown that **20–25% of early gastric cancer (EGC) cases** and **20–30% of colorectal polyps** are often missed during endoscopy.

Artificial Intelligence (AI), particularly **deep learning**, offers significant potential to overcome these limitations by improving **accuracy, consistency, and efficiency** while reducing fatigue-related errors.

This research aims to develop an **AI-assisted system** based on **YOLO (You Only Look Once)** and complementary deep learning algorithms to **eliminate blind spots** and **reduce missed lesion detection** in gastric endoscopy.

2. Research Objectives

RO1: Develop a YOLO-based AI system to identify anatomical landmarks in gastric endoscopy videos and monitor real-time mucosal coverage.

RO2: Integrate a blind-spot detection and alerting mechanism that can reduce the proportion of blind-spot-related misdiagnoses from ~16% (sensitivity ~84%) to below 5% (sensitivity >95%)

3. Literature Review & Existing Solutions

3.1 Current Solutions

- **WISENSE (Gut, 2019):** Blind-spot monitoring in EGD. Proven effective in RCT. *Limitation:* Not fully optimized for multi-device real-time deployment.
- **APT (Scientific Reports, 2025):** Automated photodocumentation with Swin Transformer. *Strength:* High accuracy (>98%). *Limitation:* Heavy architecture.
- **MMCNet (Multimedia Systems, 2024):** Lightweight CNN for landmark recognition. *Strength:* Real-time capable. *Limitation:* Limited integration with coverage tracking/alerting.

3.2 Gap Analysis

- Existing approaches focus separately on blind-spot monitoring or lesion detection.
 - Lack of an **integrated, lightweight real-time solution** that combines:
 - Landmark recognition,
 - Coverage monitoring,
 - Blind-spot alerting,
 - Automated photodocumentation.
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4. Scope of the Study

4.1 In-Scope

- Development of a YOLO-based model to detect anatomical landmarks in **static endoscopic images** from the GastroHUN dataset.
- Calculation of **mucosal coverage completeness** and blind-spot detection based on the systematic screening protocol.
- **Evaluation using the reserved 10% portion of the dataset**, with quantitative metrics: coverage completeness, blind-spot rate, sensitivity, F1-score, and inference time per image.

4.2 Out-of-Scope (Future Work)

- Extension to **video-based real-time analysis** and alerting during endoscopy procedures.
- Integration with endoscopy hardware and hospital IT systems.
- Clinical validation with live procedures and regulatory approval processes.

4.3 Expected Deliverable of the Thesis

A research prototype (**software model + evaluation results**) that demonstrates feasibility of **blind-spot reduction** in gastric endoscopy **using image-based** deep learning analysis.

5. Research Questions

RQ1: Can a YOLO-based AI system identify anatomical landmarks in gastric endoscopy videos with high precision and recall (>90%)?

RQ2: How to effectively monitor real-time mucosal coverage to ensure comprehensive inspection during gastric endoscopy?

RQ3: Does the integration of a blind-spot detection and alerting mechanism significantly reduce the rate of blind-spot-related misdiagnoses from ~16% to below 5% (sensitivity >95%)?

6. Methodology

- **Data Source:** This study uses the **GastroHUN dataset** (*Scientific Data*, 2025) – “GastroHUN: an Endoscopy Dataset of Complete Systematic Screening Protocol for the Stomach” ([DOI: 10.1038/s41597-025-04401-5](https://doi.org/10.1038/s41597-025-04401-5)). This dataset provides gastric endoscopy images and videos collected under a complete systematic screening protocol, with expert annotations.
- **Preprocessing:**
Instead of using the entire dataset, only samples **with full expert consensus**

annotations will be selected. This ensures the highest reliability of the training and evaluation data. Standard image normalization and augmentation techniques (rotation, flip, zoom, illumination adjustment) may also be applied to improve robustness, but the **core filtering step** is to rely solely on expert consensus-approved data.

- **Model:** YOLO-based landmark detection + coverage tracking module.
- **Evaluation:**
 - The dataset will be split into **90% for training** and **10% for testing**.
 - The testing subset will be used to compute quantitative metrics: coverage completeness ($\geq 95\%$), blind-spot rate ($< 5\%$), sensitivity, F1-score, and latency (ms/frame).
 - Additionally, if feasible, the **video sequences within GastroHUN** will be used for **integration testing in real-time simulation**, to validate the system's practical performance.

7. Expected Outcomes

EO1 (Primary): A YOLO-based model that can detect anatomical landmarks in gastric endoscopy images with ≥ 0.90 mAP and calculate mucosal coverage completeness $\geq 95\%$.

EO2 (Primary): Demonstrated reduction in blind-spot-related misdiagnosis proxy (missed mandatory landmarks) from $\sim 16\%$ to $< 5\%$ on the test set.

EO3 (Primary): Quantitative evidence (sensitivity, F1-score, coverage metrics) showing feasibility of the proposed system on still images.

EO4 (Deferred – Future Work): Extension to **video-based real-time integration** for alerts and workflow support (to be implemented if time/resources allow).

8. Potential Applications & Contributions

- **Clinical Impact:** Early and accurate cancer detection, improved patient outcomes, reduced misdiagnosis.
 - **Healthcare Efficiency:** Reduced workload for doctors, minimized human error, standardized procedures.
 - **Scalability:** Potential extension to colonoscopy, capsule endoscopy, bronchoscopy, and other imaging modalities.
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9. Challenges & Limitations

9.1 Limited Dataset Expansion

- Although GastroHUN provides a solid foundation, expanding the dataset to further improve model reliability is challenging.
- Medical imaging data is strictly regulated due to patient privacy and ethical concerns, making access to additional annotated data limited.
- This constraint may affect the model's generalizability across diverse patient populations and equipment types.

9.2 Real-Time Analysis Constraints

- Achieving real-time video analysis requires **high-performance and stable hardware** (e.g., powerful GPUs, optimized I/O systems).
- Without sufficient resources, system latency may exceed acceptable limits, reducing feasibility for clinical deployment.

9.3 Integration into Clinical Workflow

- Even if the technical performance is achieved, **integrating the system into existing endoscopy machines** and hospital IT infrastructure presents significant complexity.

- Additionally, obtaining **medical device certification and regulatory approval** is a lengthy and challenging process.

10. Timeline

Phase	Task	Duration
1	Literature Review & Dataset Preparation	Weeks 1–2
2	Model Development & Initial Training	Weeks 2–5
3	Validation & Optimization	Weeks 5–7
4	System Integration & Prototype	Weeks 7–8
5	Final Report Writing & Defense Preparation	Weeks 9–12

11. References

- “[Evaluation of an artificial intelligence-based system for real-time high-quality photodocumentation during esophagogastroduodenoscopy](#)” (Scientific Reports, 2025)
- “[Reducing blind spots in esophagogastroduodenoscopy examinations using a novel deep learning model](#)” (Multimedia Systems, 2024)
- “[Intelligent detection endoscopic assistant: An artificial intelligence-based system for monitoring blind spots during esophagogastroduodenoscopy in real-time](#)” (Digestive and Liver Disease, 2021)
- “[A deep neural network improves endoscopic detection of early gastric cancer without blind spots](#)” (Endoscopy, 2019)

- [“Deep Learning Models for Anatomical Location...”](#) (Diagnostics, MDPI, 2024)
- [“Deep learning for gastroscopic images: computer-aided techniques for clinicians”](#) (BioMedical Engineering OnLine, 2022)
- [“Impact of Computer-Assisted System on Learning ...”](#) (Frontiers in Medicine, 2021)